Simplicial Complex and Simplicial Homology

Pinyi Wang

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This note gives a introduction about Acyclic Model Theorem by Samuel Eilenberg and Saunders MacLane [1]. It was widely used in algebraic topology. Some simple application includes the proof of homotopy invariance of singular homology, the construction of barycentric subdivision in singular homology, and the proof of the ordered simplical homology is isomorphic to the oriented simplicial homology. Therefore, it is very important to introduce it separately.

Before the theorem, some important terminologies should be introduced.

1 Models and Expressible Functors

Let $\mathcal{C}$ be a category and $F:\mathcal{C}\to \mathrm{Ab}$ be a covarient functor. Suppose we have chosen a set $\mathfrak{M}\subseteq \mathrm{Ob}(\mathcal{C})$ and $U=\{U_M\subseteq F(M)\mid M\subset\mathfrak{M}\}$, a family of subsets of $F(M), M\in\mathfrak{M}$.

(1) We can define a functor $\widetilde{F}^U:\mathcal{C}\to\mathrm{Ab}$,

$$\widetilde{F}^U(X)=\mathbb{Z}^{\oplus\{(\phi,m)\mid\phi\in\mathrm{Hom}(M,X),m\in U_M\subseteq F(M),M\in\mathfrak{M}\}}$$

$$X\xrightarrow{f}Y, \widetilde{F}^U(X)\xrightarrow{\widetilde{F}^U(f)}\widetilde{F}^U(Y), \widetilde{F}^U(f)(\phi,m)=(f\circ\phi,m)$$

(2) We have a natural transformation $\widetilde{F}^U\xrightarrow{\pi}F$

$$\widetilde{F}^U(X)\xrightarrow{\pi(X)}F(X)$$

$$\pi(X)(\phi,m)=F(\phi)m$$

(3) We say F is $U-$expressible if π is a natural equivalence. Therefore, if F is $U-$expressible, then $F(X)$ is a **free abelian group** due to the isomorphism $\pi(X)$ for all $X\in\mathrm{Ob}(\mathcal{C})$ and a basis is $\{F(\phi)m\mid\phi:M\to X,m\in U_M\subseteq F(M)\}$. Conversely, if $F(X)$ is freely generated by $\{F(\phi)m\mid\phi:M\to X,m\in U_M\subseteq F(M)\}$, then we can construct a natural transformation $\tau:F\to\widetilde{F}^U$ and prove that it is a natural equivalence. Therefore, F is $U-$expressible if and only if $F(X)$ is freely generated by $\{F(\phi)m\mid\phi:M\to X,m\in U_M\subseteq F(M)\}$.

Example 1.1. *Consider the category of topological spaces $\mathcal{C} = \text{Top}$. Let $F = S_k$ be the functor of singular chain group. $\mathfrak{M} = \{\Delta_k\}$ where Δ_k is the standard k-dimensional simplex. $U_{\Delta_k} = \{id_{\Delta_k}\} \subseteq S_k(\Delta_k)$. Then S_k is $U-$expressible. The basis of $\widetilde{S}_k(X)$ is $\{(\phi, id_{\Delta_k}) \mid \phi : \Delta_k \to X\}$. The basis of $S_k(X)$ is $\{\phi : \Delta_k \to X\}$. $\pi(X)(\phi, id_{\Delta_k}) = S_k(\phi) id_{\Delta_k} = \phi$, so $\pi(X)$ is a one-to-one correspondence between two basis, so $\pi(X)$ is an isomorphism and S_k is $U-$expressible.*

2 Acyclic Model Theorems

Theorem 2.1. *Suppose we have covariant functors from $\mathcal{C}$ to* Ab *and natural transformations between them*

$$\begin{tikzcd}
 & K_k \arrow[r, "\partial_k"] \arrow[d, "f_k"] & K_{k-1} \arrow[r, "\partial_{k-1}"] \arrow[d, "f_{k-1}"] \arrow[dl, "h_{k-1}"] & K_{k-2} \arrow[dl, "h_{k-2}"] \\
K'_{k+1} \arrow[r, "\partial'_{k+1}"] & K'_k \arrow[r, "\partial'_k"] & K'_{k-1} &
\end{tikzcd}$$

such that $\partial_k \circ \partial_{k-1} = 0$, $\partial'_{k+1} \circ \partial'_k = 0$, $f_{k_1} \circ \partial_k = \partial'_k \circ f_k$ and $f_{k-1} = h_{k-2} \circ \partial_{k-1} + \partial'_k \circ h_{k-1}$. If K_k is $U-$expressible and $\text{Ker } \partial'_k(M) = \text{Im } \partial'_{k+1}, \forall M \in \mathfrak{M}$, *then there exists a natural tranformaiton $h_k : K_k \to K'_{k+1}$ such that $f_k = \partial'_{k+1} \circ h_k + h_{k-1} \circ \partial_k$.*

Remark 2.1. *All the elements in $\mathfrak{M}$ are called models. For each model $M \in \mathfrak{M}$, the condition* $\text{Ker } \partial'_k(M) = \text{Im } \partial'_{k+1}, \forall M \in \mathfrak{M}$ *means that the sequence is exact at K'_k for all models. If a chain complex has trivial homology then we call it acyclic. That explains where the name of this theorem comes from.*

Proof. Step 1: Define $h_k(M)u$ for $u \in U_M \subseteq K_k(M)$, $M \in \mathfrak{M}$. Consider the following diagram and $u \in U_M \subseteq K_k(M)$.

$$\begin{tikzcd}
 & u \in K_k(M) \arrow[r, "\partial_k(M)"] \arrow[d, "f_k(M)"] & K_{k-1}(M) \arrow[r, "\partial_{k-1}(M)"] \arrow[d, "f_{k-1}(M)"] \arrow[dl, "h_{k-1}(M)"] & K_{k-2}(M) \arrow[dl, "h_{k-2}(M)"] \\
K'_{k+1}(M) \arrow[r, "\partial'_{k+1}(M)"] & K'_k(M) \arrow[r, "\partial'_k(M)"] & K'_{k-1}(M) &
\end{tikzcd}$$

We want to find $h_k(M)u \in K'_{k+1}(M)$ such that $f_k(M)u = h_{k-1}(M) \circ \partial_k(M)u + \partial'_{k+1}(M) \circ h_k(M)u$. Note that $f_k(M)u - h_{k-1}(M) \circ \partial_k(M)u \in \text{Ker } \partial'_k(M)$, so $f_k(M)u - h_{k-1}(M) \circ \partial_k(M)u \in \text{Im } \partial'_{k+1}(M)$. Choose $\xi_u \in K'_{k+1}(M)$ such that $\partial'_{k+1}(M)\xi_u = f_k(M)u - h_{k-1}(M) \circ \partial_k(M)u$.

Step 2: Define a natural transformation $\widetilde{K}_k^U \xrightarrow{\widetilde{h}_k} K'_{k+1}$: $\forall X \in \mathrm{Ob}(\mathcal{C})$, $\widetilde{K}_k^U(X)$ is a free abelian group which basis has the form (ϕ, u) for $\phi : M \to X$ and $u \in U_M$. Then let $\widetilde{h}_k(X)(\phi, m) = K'_{k+1}(\phi)\xi_u$. $\widetilde{h}_k$ is a natural transformation because the following diagram commutes:

$$\begin{tikzcd}
(\phi, u) \in \widetilde{K}_k^U(X) \arrow[r, "\widetilde{h}_k(X)"] \arrow[d, "\widetilde{K}_k^U(f)"] & K'_{k+1}(\phi)\xi_u \in K'_{k+1}(X) \arrow[d, "K'_{k+1}(f)"] \\
(f \circ \phi, u) \in \widetilde{K}_k^U(Y) \arrow[r, "\widetilde{h}_k(Y)"] & K'_{k+1}(f \circ \phi)\xi_u \in K'_{k+1}(Y)
\end{tikzcd}$$

for all $f : X \to Y$.

Step 3: We have $f_k \circ \pi = \partial'_{k+1} \circ \widetilde{h}_{k+1} + h_{k-1} \circ \partial_k \circ \pi$. That is to say, we have the following commutative diagram:

$$\begin{tikzcd}
 & \widetilde{K}_k^U \arrow[dl, "\widetilde{h}_k"'] \arrow[d, "\pi"] & \\
 & K_k \arrow[r, "\partial_k"] \arrow[d, "f_k"] & K_{k-1} \arrow[dl, "h_{k-1}"] \\
K'_{k+1} \arrow[r, "\partial'_{k+1}"] & K'_k &
\end{tikzcd}$$

Therefore, for each (ϕ, u), $\phi : M \to X$, $u \in U_M \subseteq K_k(M)$, we have the following commutative diagram:

$$\begin{tikzcd}
 & & K_k(M) \arrow[rr, "\partial_k(M)"] \arrow[dr, "K_k(\phi)"] \arrow[ddr, "f_k(M)"] & & K_{k-1}(M) \arrow[dr, "K_{k-1}(\phi)"] \arrow[ddl, "h_{k-1}(M)"] & \\
\widetilde{K_k}^U(M) \arrow[urr, "\pi(M)"] \arrow[dr, "\widetilde{K_k}^U(\phi)"] \arrow[ddr, dashed, "\widetilde{h_k}(M)"'] & & & K_k(X) \arrow[rr, "\partial_k(X)"] \arrow[ddr, "f_k(X)"] & & K_{k-1}(X) \arrow[ddl, "h_{k-1}(X)"] \\
 & \widetilde{K_k}^U(X) \arrow[urr, "\pi(X)"] \arrow[ddr, dashed, "\widetilde{h_k}(X)"] & & K'_k(M) \arrow[dr, "K'_k(\phi)"] & & \\
 & K'_{k+1}(M) \arrow[urr, "\partial'_{k+1}(M)"] \arrow[dr, "K'_{k+1}(\phi)"'] & & & K'_k(X) & \\
 & & K'_{k+1}(X) \arrow[urr, "\partial'_{k+1}(X)"] & & &
\end{tikzcd}$$

Therefore,

$$\begin{aligned}
&f_k(X) \circ \pi(X)(\phi, u) \\
=&f_k(X) \circ K_k(\phi)(u) \\
=&K'_k(\phi) \circ f_k(M)(u) \\
=&K'_k(\phi)(\partial'_{k+1}\xi_u + h_{k-1}(M) \circ \partial_k(M)(u)) \\
=&\partial'_{k+1}(X) \circ K'_{k+1}(\phi)\xi_u + K'_k(\phi) \circ h_{k-1}(M) \circ \partial_k(M)u \\
=&\partial'_{k+1}(X) \circ \widetilde{h}_k(X)(\phi, u) + K'_k(\phi) \circ h_{k-1}(M) \circ \partial_k(M)u \\
=&\partial'_{k+1}(X) \circ \widetilde{h}_k(X)(\phi, u) + h_{k-1}(X) \circ \partial_k(X) \circ \pi(X)(\phi, u)
\end{aligned}$$

Therefore, $\widetilde{h}_k \circ \pi^{-1}$ is the natural transformation we are looking for. □

Remark 2.2. *In fact, π does not necessarly need to be a natural equivalence. The theorem still holds if π has a right natural inverse. This induces another concept: we say F is weakly $U-$expressible if there exists a natural transformation $\tau : F \to \widetilde{F}^U$ such that $\pi \circ \tau = id$.*

Similar to Theorem 2.1, we have the following theorem:

Theorem 2.2. *Suppose we have covariant functors from $\mathcal{C}$ to* Ab *and natural transformations between them*

$$\begin{tikzcd}
K_k \arrow[r, "\partial_k"] & K_{k-1} \arrow[r, "\partial_{k-1}"] \arrow[d, "f_{k-1}"] & K_{k-2} \arrow[d, "f_{k-2}"] \\
K'_k \arrow[r, "\partial'_k"] & K'_{k-1} \arrow[r, "\partial'_{k-1}"] & K'_{k-2}
\end{tikzcd}$$

such that $\partial_k \circ \partial_{k-1} = 0$, $\partial'_k \circ \partial'_{k-1} = 0$, $f_{k-2} \circ \partial_{k-1} = \partial'_{k-1} \circ f_{k-1}$. If K_k is weakly $U-$expressible and $\text{Ker } \partial'_{k-1}(M) = \text{Im } \partial'_k, \forall M \in \mathfrak{M}$*, then there exists a natural tranformaiton $f_k : K_k \to K'_k$ such that $f_{k-1} \circ \partial_k = \partial'_k \circ f_k$.*

The proof is exactly the same as Theorem 2.1.

References

[1] Samuel Eilenberg and Saunders MacLane, Acyclic models, Amer. J. Math. 75 (1953), 189–199.